Name: Sagar Ganesh Sharma
Employee ID : 102033

ISO Messages Assignment

- **Hexadecimal to Binary Conversion**

Hexadecimal	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

- **Convert below bitmaps into data elements**
  - **F67B C681 28E1 9010 0000 000C 0000 0000**
    - **Hexadecimal to Binary Conversion**

F = 1111 6 = 0110 7 = 0111 B = 1011	C = 1100 6 = 0110 8 = 1000 1 = 0001	2 = 0010 8 = 1000 E = 1110 1 = 0001	9 = 1001 0 = 0000 1 = 0001 0 = 0000	0 = 0000 0 = 0000 0 = 0000 0 = 0000	0 = 0000 0 = 0000 0 = 0000 C = 1100	0 = 0000 0 = 0000 0 = 0000 0 = 0000	0 = 0000 0 = 0000 0 = 0000 0 = 0000

    - **Combined Binary**
      1111 0110 0111 1011 1100 0110 1000 0001 0010 1000 1110 0001 1001 0000 0001 0000
      0000 0000 0000 0000 0000 0000 0000 1100 0000 0000 0000 0000 0000 0000 0000 0000
    - **Identifying Data Elements**
      - **DE:** 1 (Secondary Bitmap Present)
      - DE: 2, 3, 4, 6, 7, 10, 11, 12, 13, 15, 16, 17, 18, 22, 23, 25, 32, 35, 37, 41, 42, 43, 48, 49, 52, 60
      - DE: 93, 94
  - **F030 C101 0EA0 8118 0000 0000 1400 002C**
    - **Hexadecimal to Binary Conversion**

F = 1111 0 = 0000 3 = 0011 0 = 0000	C = 1100 1 = 0001 0 = 0000 1 = 0001	0 = 0000 E = 1110 A = 1010 0 = 0000	8 = 1000 1 = 0001 1 = 0001 8 = 1000	0 = 0000 0 = 0000 0 = 0000 0 = 0000	0 = 0000 0 = 0000 0 = 0000 0 = 0000	1 = 0001 4 = 0100 0 = 0000 0 = 0000	0 = 0000 0 = 0000 2 = 0010 C = 1100

    - **Combined Binary**
      1111 0000 0011 0000 1100 0001 0000 0001 0000 1110 1010 0000 1000 0001 0001 1000
      0000 0000 0000 0000 0000 0000 0000 0000 0001 0100 0000 0000 0000 0000 0010 1100
    - **Identifying Data Elements**
      - DE: 1 (Secondary Bitmap Present)

            - DE: 2, 3, 4, 11, 12, 17, 18, 24, 32, 37, 38, 39, 41, 43, 49, 56, 60, 61
            - DE: 100, 102, 123, 125, 126
    - **0000 0000 1400 002C 0000 0000 0000 0000**
        - **Hexadecimal to Binary Conversion**

0 = 0000 0 = 0000 0 = 0000 0 = 0000	0 = 0000 0 = 0000 0 = 0000 0 = 0000	1 = 0001 4 = 0100 0 = 0000 0 = 0000	0 = 0000 0 = 0000 2 = 0010 C = 1100	0 = 0000 0 = 0000 0 = 0000 0 = 0000	0 = 0000 0 = 0000 0 = 0000 0 = 0000	0 = 0000 0 = 0000 0 = 0000 0 = 0000	0 = 0000 0 = 0000 0 = 0000 0 = 0000

        - **Combined Binary**
            - 0000 0000 0000 0000 0000 0000 0000 0000 0001 0100 0000 0000 0000 0000 0010 1100 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
        - **Identifying Data Elements**
            - DE: 36, 38, 59, 61, 62

- **Convert below values into bitmaps**
    - 1, 2, 3, 4, 11, 12, 17, 24, 32, 37, 39, 41, 43, 48, 49, 102, 123, 125
        - **Binary Data**
            - 1111 0000 0011 0000 1000 0001 0000 0001 0000 1010 1010 0001 1000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0100 0000 0000 0000 0000 0010 1000
        - **Binary to Hexadecimal**

1111 = F 0000 = 0 0011 = 3 0000 = 0	1000 = 8 0001 = 1 0000 = 0 0001 = 1	0000 = 0 1010 = A 1010 = A 0001 = 1	1000 = 8 0000 = 0 0000 = 0 0000 = 0	0000 = 0 0000 = 0 0000 = 0 0000 = 0	0000 = 0 0000 = 0 0000 = 0 0000 = 0	0000 = 0 0100 = 4 0000 = 0 0000 = 0	0000 = 0 0000 = 0 0010 = 2 1000 = 8

        - **Bitmap:** F030 8101 0AA1 8000 0000 0000 0400 0028

    - 2, 3, 4, 11, 12, 17, 24, 32, 37, 39, 41, 43, 48, 49, 51, 55
        - **Binary Data:**
            - 0111 0000 0011 0000 1000 0001 0000 0001 0000 1010 1010 0001 1010 0010 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000
        - **Binary to Hexadecimal**

0111 = 7 0000 = 0 0011 = 3 0000 = 0	1000 = 8 0001 = 1 0000 = 0 0001 = 1	0000 = 0 1010 = A 1010 = A 0001 = 1	1010 = A 0010 = 2 0000 = 0 0000 = 0	0000 = 0 0000 = 0 0000 = 0 0000 = 0	0000 = 0 0000 = 0 0000 = 0 0000 = 0	0000 = 0 0000 = 0 0000 = 0 0000 = 0	0000 = 0 0000 = 0 0000 = 0 0000 = 0

        - **Bitmap:** 7030 8101 0AA1 A200 0000 0000 0000 0000

- **Breakup for below MTI.**
  - **1804**
    - 1 → ISO 8583:1987 version
    - 8 → Network Management Message
    - 0 → Response
    - 4 → Reserved for future use
  - **0221**
    - 0 → ISO 8583:1987 version
    - 2 → Financial Transaction
    - 2 → Advice Repeat
    - 1 → Response
  - **0420**
    - 0 → ISO 8583:1987 version
    - 4 → Reversal/Chargeback
    - 2 → Advice
    - 0 → Request
  - **0800**
    - 0 → ISO 8583:1987 version
    - 8 → Network Management
    - 0 → Request
    - 0 → Original
  - **0100**
    - 0 → ISO 8583:1987 version
    - 1 → Authorization
    - 0 → Request
    - 0 → Original
  - **9906 : Account Data Selection**
    - 9 → ATM/PoS Related
    - 9 → Proprietary of ATM/PoS
    - 0 → Input
    - 6 → Select Account

- **Self study:**
- **What is tertiary bitmaps?**
  - Primary Bitmap (DE 1) – Indicates the presence of Data Elements 1–64
  - Secondary Bitmap (bit 1 of the primary bitmap = 1) – Indicates Data Elements 65–128
  - Tertiary Bitmap (bit 1 of the secondary bitmap = 1) – Indicates Data Elements 129–192
  - It's a third set of 64 bits, indicating presence of DE129–DE192
  - Only used in extended/custom implementations, not in the original ISO 8583:1987
  - Tertiary bitmap exists only if bit 1 of the secondary bitmap is set to 1
  - It's very rare, used in highly customized systems, or financial networks needing very specific extended data elements
  - ISO 8583:1987 standard only defines up to DE128
  - Tertiary bitmap is a non-standard extension (used in ISO 8583:2003 or custom implementations)
  - Most standard systems (e.g., BASE24, Postilion, etc.) do not use tertiary bitmap unless they support the 2003 version or a custom schema